

Figure 1

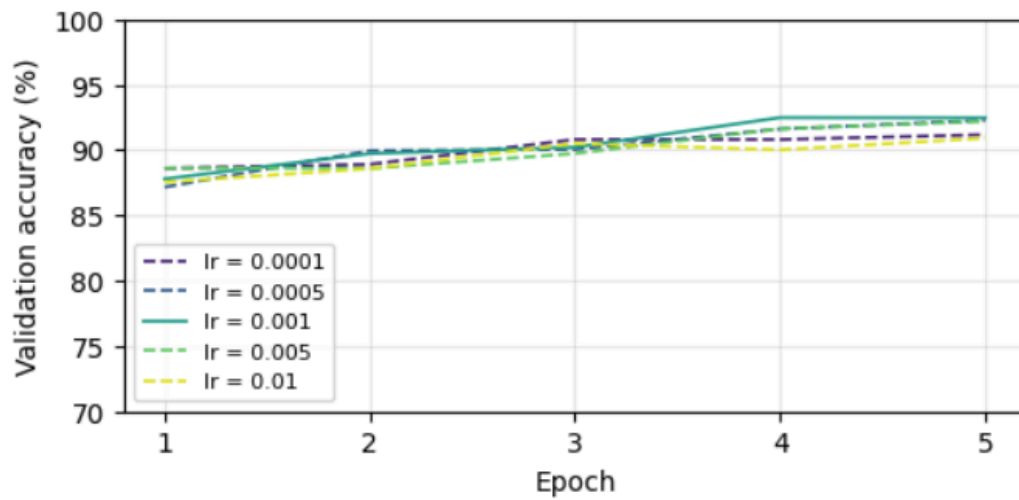


Figure 1. Validation accuracy over 5 epochs for the BatchNorm CNN trained with five Adam LR. The solid line marks the best LR ($lr = 0.001$), selected based on highest validation accuracy at epoch 5.

Training: Baseline (no BN) lr=0.001

epoch 1/15	train 0.5939 / 77.8%	val 0.3836 / 86.0%
epoch 2/15	train 0.3298 / 87.8%	val 0.2793 / 89.4%
epoch 3/15	train 0.2696 / 90.2%	val 0.2675 / 90.4%
epoch 4/15	train 0.2340 / 91.4%	val 0.2414 / 91.0%
epoch 5/15	train 0.2096 / 92.3%	val 0.2170 / 91.9%
epoch 6/15	train 0.1850 / 93.1%	val 0.2453 / 91.0%
epoch 7/15	train 0.1649 / 93.8%	val 0.2052 / 92.4%
epoch 8/15	train 0.1461 / 94.6%	val 0.2182 / 92.1%
epoch 9/15	train 0.1284 / 95.1%	val 0.2322 / 92.4%
epoch 10/15	train 0.1080 / 96.0%	val 0.2584 / 92.1%
epoch 11/15	train 0.0908 / 96.6%	val 0.2537 / 92.3%
epoch 12/15	train 0.0796 / 96.9%	val 0.2547 / 92.3%
epoch 13/15	train 0.0683 / 97.5%	val 0.2800 / 92.5%
epoch 14/15	train 0.0613 / 97.7%	val 0.2937 / 92.4%
epoch 15/15	train 0.0551 / 98.0%	val 0.3255 / 91.8%

Training: BatchNorm lr=0.001

epoch 1/15	train 0.3882 / 85.8%	val 0.3260 / 87.8%
epoch 2/15	train 0.2624 / 90.3%	val 0.2836 / 89.7%
epoch 3/15	train 0.2208 / 91.9%	val 0.2686 / 90.2%
epoch 4/15	train 0.1902 / 92.9%	val 0.2102 / 92.5%
epoch 5/15	train 0.1682 / 93.8%	val 0.2159 / 92.5%
epoch 6/15	train 0.1438 / 94.6%	val 0.2544 / 91.6%

...

epoch 12/15	train 0.0557 / 97.9%	val 0.3030 / 92.4%
epoch 13/15	train 0.0457 / 98.3%	val 0.3253 / 92.4%
epoch 14/15	train 0.0424 / 98.5%	val 0.2844 / 92.8%
epoch 15/15	train 0.0373 / 98.6%	val 0.3598 / 92.2%

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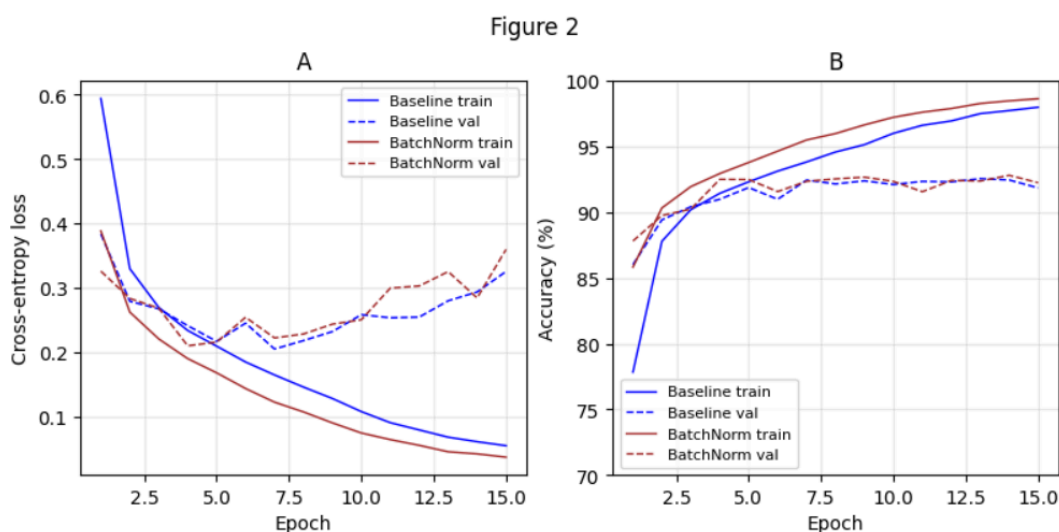


Figure 2. Training (solid line) and validation (dashed line) metrics over 15 epochs for the baseline CNN without BatchNorm (blue) and the BatchNorm CNN (brown), both trained at lr = 0.001. A) Cross-entropy loss. B) Classification accuracy (%).

Test-set results:

Baseline CNN loss: 0.3571 accuracy: 91.63%

BatchNorm CNN loss: 0.3937 accuracy: 91.60%

Per-class test accuracy (BatchNorm CNN):

T-shirt/top	: 92.6%
Trouser	: 98.9%
Pullover	: 90.1%
Dress	: 93.7%
Coat	: 87.1%
Sandal	: 96.1%
Shirt	: 65.5%
Sneaker	: 98.9%
Bag	: 98.1%
Ankle boot	: 95.0%

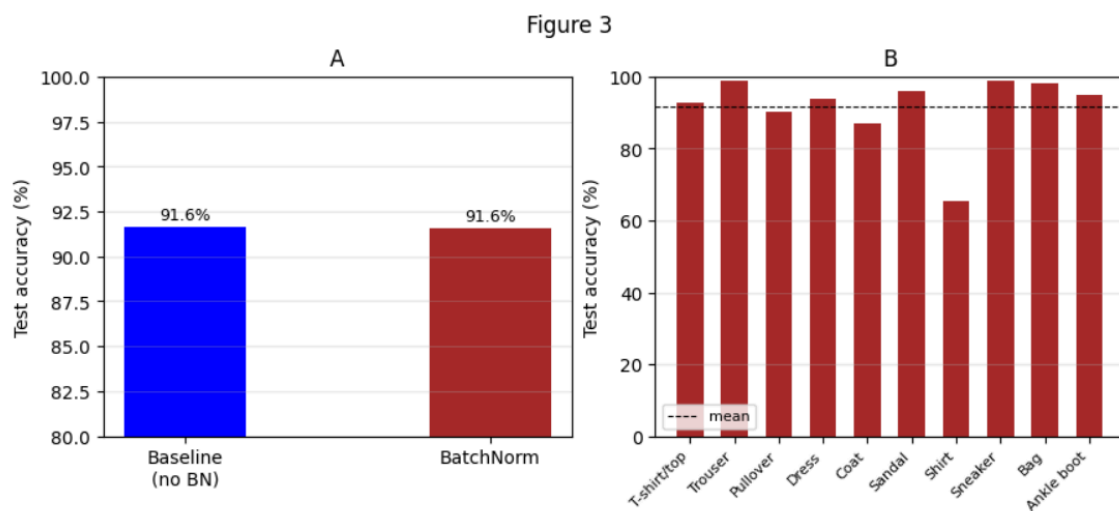


Figure 3. Test set performance of the two CNN variants, with and without BN. A) Overall test accuracy for the baseline (without BatchNorm, 91.6%) and BatchNorm CNN (91.6%), showing comparable test-set performance. B) Per class accuracy for the BatchNorm model, overall mean represented by top dash line. Shirt is consistently the lower performance class due to visual overlap with T-shirt, Pullover and Coat.

Figure 4

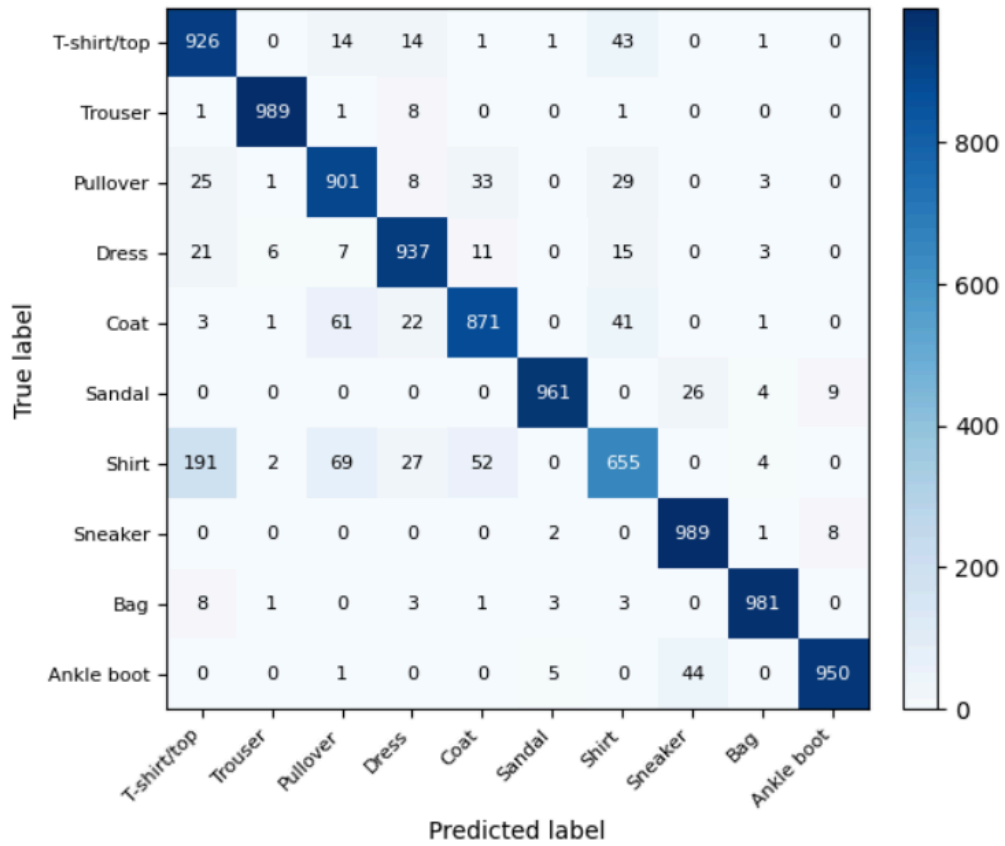


Figure 4. 10x10 Confusion matrix for the BatchNorm CNN on the 10000 image test set. Rows are the true class, columns are the predicted class. Diagonal entries are correct predictions. The largest off-diagonal counts involve Shirt, which is confused with T-shirt/top, Pullover, and Coat garments.

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Training GAN for 30 epochs on cpu...
epoch 1/30 D loss: 0.1675 G loss: 2.5297
epoch 2/30 D loss: 0.2273 G loss: 2.3063
epoch 3/30 D loss: 0.2888 G loss: 2.0127
epoch 4/30 D loss: 0.3121 G loss: 1.9485
epoch 5/30 D loss: 0.3114 G loss: 1.9801
epoch 6/30 D loss: 0.3038 G loss: 2.0352
epoch 7/30 D loss: 0.3006 G loss: 2.0843
epoch 8/30 D loss: 0.2912 G loss: 2.1437
epoch 9/30 D loss: 0.2879 G loss: 2.1903
epoch 10/30 D loss: 0.2825 G loss: 2.2317
    → Epoch 10 classifier check: top class = T-shirt/top (14.0%), mean max-prob = 0.896
epoch 11/30 D loss: 0.2811 G loss: 2.2579
epoch 12/30 D loss: 0.2750 G loss: 2.3048
epoch 13/30 D loss: 0.2693 G loss: 2.3475
epoch 14/30 D loss: 0.2721 G loss: 2.3714
epoch 15/30 D loss: 0.2715 G loss: 2.3951
epoch 16/30 D loss: 0.2653 G loss: 2.4140
epoch 17/30 D loss: 0.2649 G loss: 2.4476
epoch 18/30 D loss: 0.2613 G loss: 2.4641
epoch 19/30 D loss: 0.2609 G loss: 2.4908
epoch 20/30 D loss: 0.2579 G loss: 2.5278
    → Epoch 20 classifier check: top class = T-shirt/top (16.2%), mean max-prob = 0.919
epoch 21/30 D loss: 0.2502 G loss: 2.5437
epoch 22/30 D loss: 0.2521 G loss: 2.5930
...
epoch 30/30 D loss: 0.2367 G loss: 2.7427
    → Epoch 30 classifier check: top class = T-shirt/top (16.6%), mean max-prob = 0.900

GAN training finished

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Figure 5

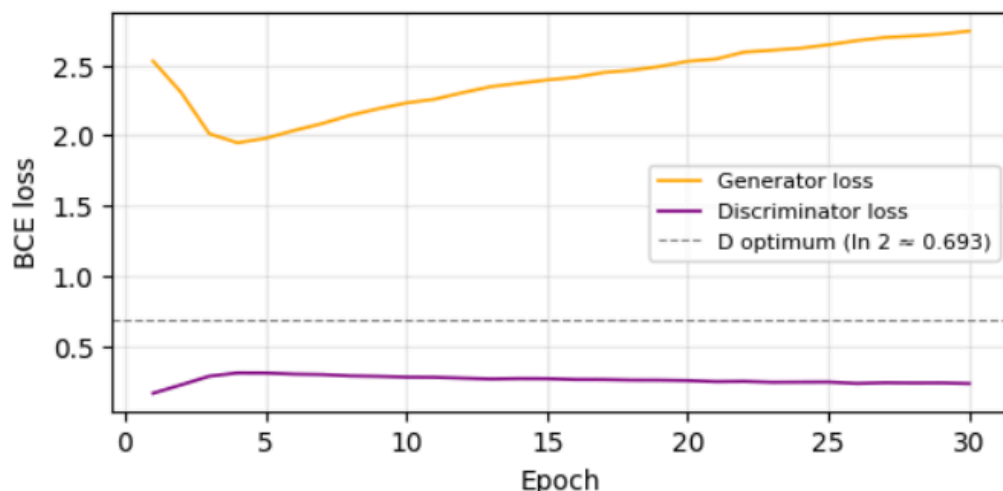


Figure 5. Generator (orange) and Discriminator (purple) BCE loss over 30 epochs of GAN training on Fashion MNIST. The dashed grey line marks the theoretical Discriminator optimum ($\ln 2 \approx 0.693$) at which D cannot distinguish real image from generated images

Figure 6



Figure 6. Grid of 64 images generated by the trained Generator using a fixed set of noise vectors $z \sim N(0,1)$. All 64 images were produced from the same z vectors after 30 epochs of training, so quality reflects the final state of the Generator. Images show recognisable Fashion MNIST clothing shapes but with some blurring and artefacts typical of GAN-generated images.

Real images, mean D score: 0.628 (std: 0.257)
Fake images, mean D score: 0.144 (std: 0.149)

Figure 7

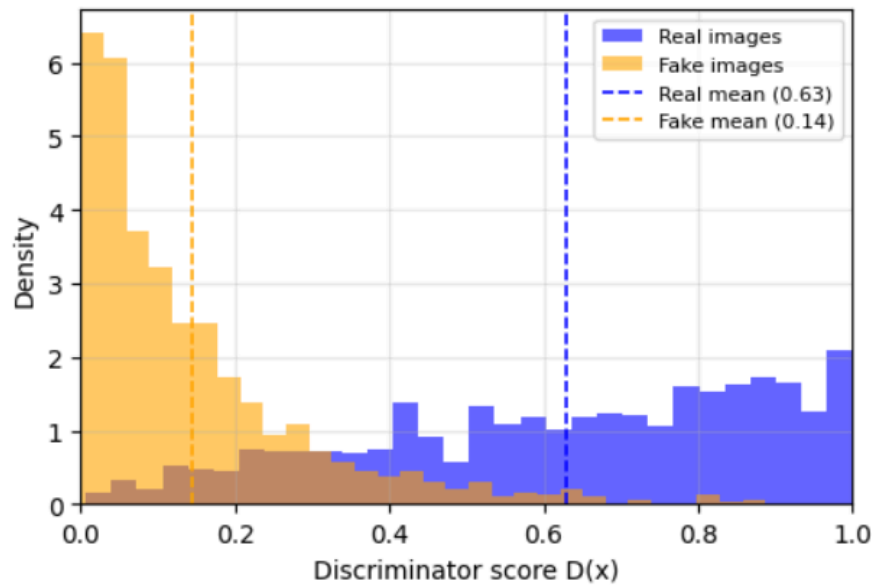


Figure 7. Normalised histogram of Discriminator scores $D(x)$ for 1000 real (blue) and 1000 generated (orange) Fashion MNIST images after GAN training. Dashed lines mark the mean score for each group. Separation between the two distributions indicates the degree to which the Discriminator can still distinguish real from generated images.

Figure 8

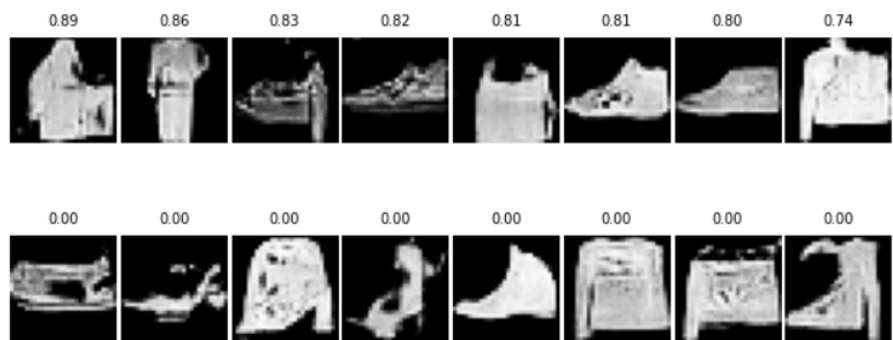


Figure 8 generated images ranked by discriminator score $D(x)$. Top row: 8 generated images with the highest D score, the fakes that best convinced the discriminator they were real, labelled with their score. Bottom row: 8 generated images with the lowest D score, those most easily identified as fake. Higher-scoring images tend to show cleaner structure and more realistic textures than lower-scoring ones.

Figure 9

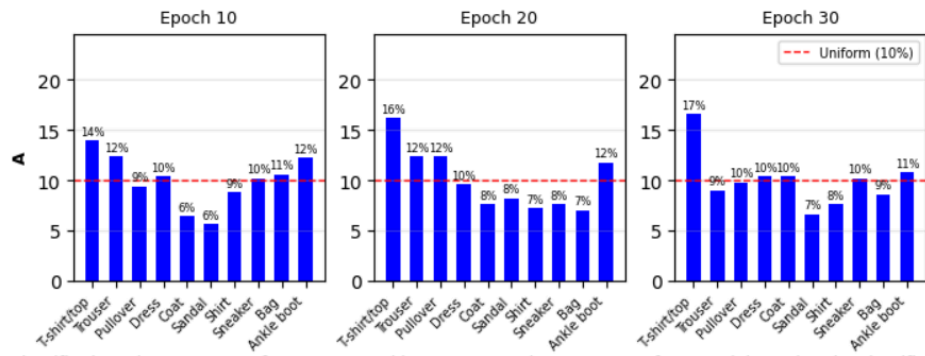


Figure 9. Classifier-based assessment of 500 generated images at epochs 10, 20, 30 of GAN training using the classifier from task 1. figure row A, the bars show the percentage of generated images assigned to each Fashion MNIST class; the red dashed line marks the 10% uniform expectation and deviation indicates partial mode collapse.

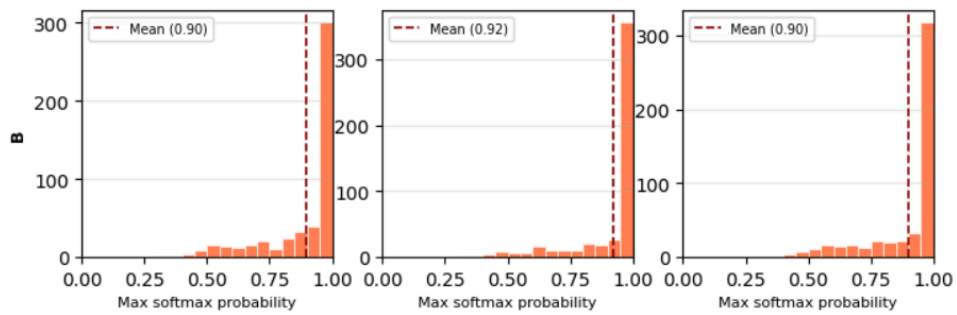


figure row B, the histograms of the maximum softmax probability per generated image. A peak near 1.0 indicates the classifier confidently assigns images to a single class; low max-probability values indicate spread predictions, typically from blurry or incoherent images.

Figure 13

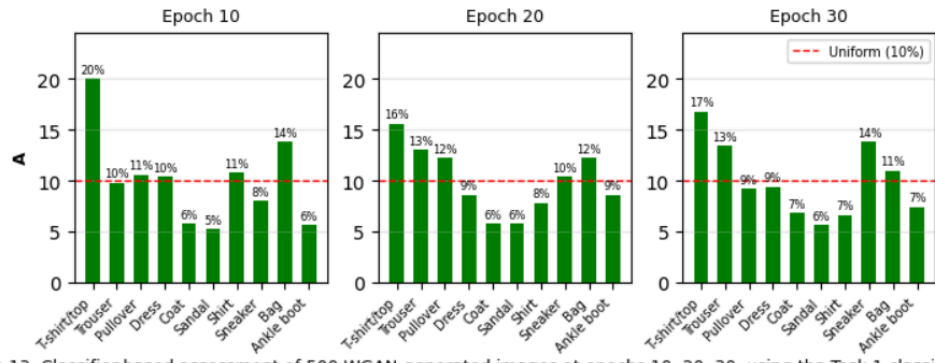
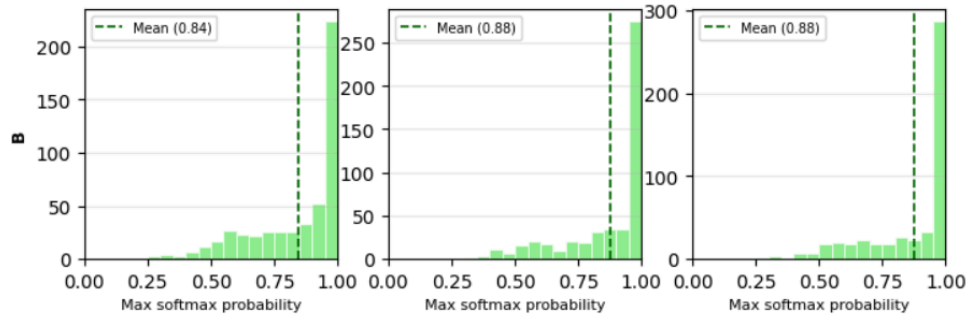


Figure 13. Classifier-based assessment of 500 WGAN-generated images at epochs 10, 20, 30, using the Task 1 classifier. Row A: percentage of images assigned to each Fashion-MNIST class; the dashed line marks the 10% uniform expectation.



Row B: histograms of the maximum softmax probability per WGAN-generated image. A peak near 1.0 indicates the classifier confidently assigns images to a single class

WGAN Real images, mean critic score: -0.042 (std: 0.009)
WGAN Fake images, mean critic score: -0.049 (std: 0.010)

Figure 12



Figure 12. Normalised histogram of WGAN critic scores $C(x)$ for 1000 real (blue) and 1000 generated (orange) Fashion-MNIST images after training. Higher scores indicate images rated as more real by the critic. Separation between the distributions indicates the degree to which the critic can still distinguish real from generated images after training.

Figure 11



Figure 11. Grid of 64 images generated by the WGAN Generator from the same fixed noise vectors as Figure 6, after 30 epochs of Wasserstein training. Using identical noise vectors allows direct visual comparison of image quality between the minimax GAN (Figure 6) and the WGAN (Figure 11).

Figure 10

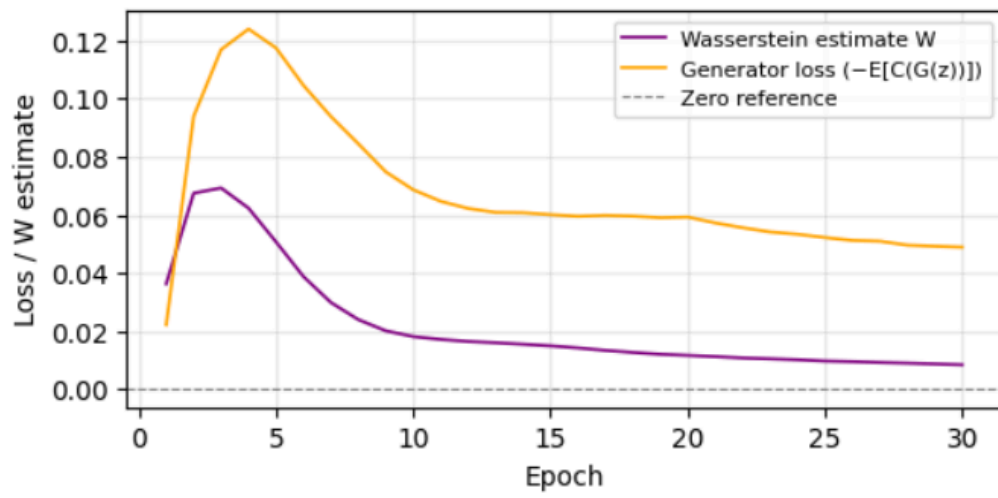


Figure 10. WGAN training curves over 30 epochs. Purple: Wasserstein estimate $W = E[C(\text{real})] - E[C(\text{fake})]$, the critic's approximation of the distance between the real and generated distributions. Orange: Generator loss $-E[C(G(z))]$.